HOCHSCHULE
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OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 1: Basics

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1 Basics

1.1 Quantities, Units and Ohm's Law

This chapter collects the tools that every later chapter uses: the basic quantities and their units, Ohm's law, Kirchhoff's laws, the voltage divider, equivalent sources, the LED with its series resistor and the RC circuit. Most of it is known from the introductory course on electrical engineering. Here it is repeated with a view to the lab: at the end of the semester you will build a measurement chain "From Sensor to Digital Value". A resistive sensor sits in a bridge, an instrumentation amplifier scales the signal to $0\,\mathrm{V} \ldots 4\,\mathrm{V}$, a low-pass filter limits the bandwidth, a sample and hold circuit freezes the value and a 3-bit flash converter with a resistor ladder turns it into a number that is shown on LEDs. Every block of this chain is built from the elements of this chapter.

We also introduce the circuit simulator Falstad (CircuitJS). Every circuit of the lecture exists as a Falstad file, so you can change values and watch the effect immediately.

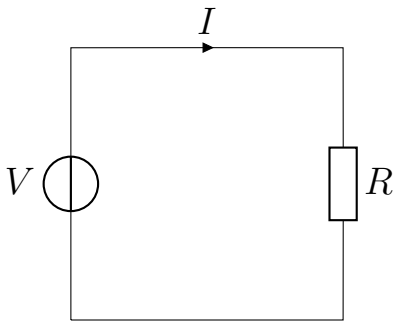
Quantities and units

quantity	symbol	unit	definition
voltage	V	volt, V	energy per charge, J/C
current	I	ampere, A	charge per time, C/s
resistance	R	ohm, Ω	$R = V/I$
power	P	watt, W	$P = V \cdot I$
capacitance	C	farad, F	$C = Q/V$
time, frequency	t, f	s, Hz	$f = 1/T$

- prefixes: p (10^{-12}), n (10^{-9}), µ (10^{-6}), m (10^{-3}), k (10^{3}), M (10^{6})
- useful combination: V / kΩ = mA, mA · kΩ = V

Calculating with kΩ, mA and V avoids most errors with powers of ten: 5 V across 2.2 kΩ gives 2.27 mA without any exponent. Voltages are always measured *between two points*. If only one point is named, the second one is ground (0 V). In the diagrams voltages are drawn as arrows, currents as arrows on the wires; the direction of an arrow defines the sign, not the physics. A negative result simply means that the real direction is opposite to the arrow.

Ohm's law



Ohm's law

$$V = R \cdot I \qquad I = \frac{V}{R} \qquad R = \frac{V}{I}$$

- a resistor is **linear**: double voltage ⇒ double current
- the conductance $G = 1/R$ is measured in siemens, S

Ohm's law describes resistors, but many other components are nonlinear: for a diode or an LED the current grows exponentially with the voltage, so there is no single value R. For such components we use a characteristic curve or a simplified model, for example a constant forward voltage V_F.

Power and power rating

Power in a resistor

$$P = V \cdot I = I^2\,R = \frac{V^2}{R}$$

- the power is turned into heat; every resistor has a **power rating**
  - SMD 0603: 0.1 W, SMD 1206: 0.25 W, through-hole (0207): 0.25 W … 0.6 W
- design rule: stay below about $50\,\%$ of the rating
- small signal circuits (kΩ, mA) dissipate only mW
- example: $100\,\Omega$ at 5 V: $P = 25\,\text{V}^2/100\,\Omega = 250$ mW, too much for a 0.25 W resistor

The power rating is specified at an ambient temperature of typically $70\,^\circ$C; above that it must be derated. A resistor at its full rating becomes hot (well above $100\,^\circ$C), its value drifts and its life is shortened. Therefore we keep a safety factor of two. In measurement circuits the dissipated power matters for another reason: a sensor that heats itself measures its own temperature. This *self-heating* is the reason why the sensor current in the capstone task is limited to 1 mA.

Standard values: the E-series

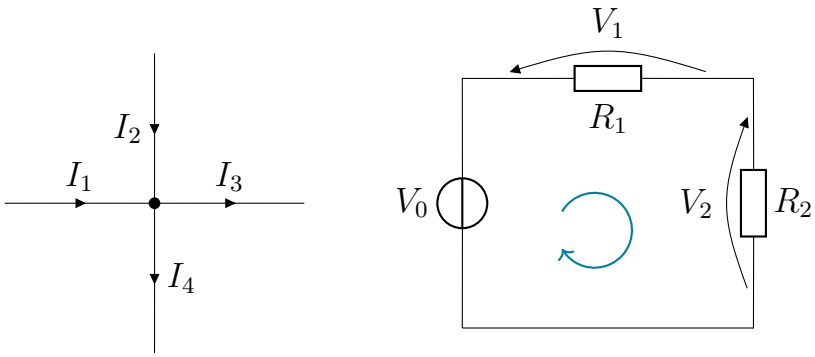
- resistors are not available in every value, but in logarithmic series
- E12 ($10\,\%$ tolerance): 1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2
- E24 ($5\,\%$) adds: 1.1 1.3 1.6 2.0 2.4 3.0 3.6 4.3 5.1 6.2 7.5 9.1
- E96 ($1\,\%$): 96 values per decade, standard for metal film resistors today
- each value exists in every decade: $4.7\,\Omega$, $47\,\Omega$, …, 4.7 MΩ

The neighbouring values of a series differ by a constant factor ($10^{1/12} \approx 1.21$ for E12), so that the tolerance bands of neighbouring values just touch. Today most resistors have a tolerance of $1\,\%$ even if one uses E12 or E24 values; the coarser series are simply easier to remember and to keep in stock. A design normally ends with choosing the next standard value and recalculating the result with it. Capacitors are usually available in E6 (1.0, 1.5, 2.2, 3.3, 4.7, 6.8).

1.2 Kirchhoff's Laws and Networks

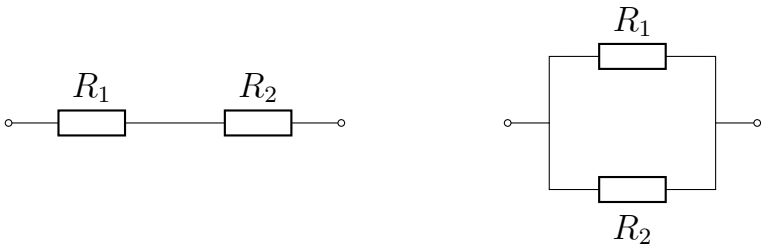
Two conservation laws are enough to analyse any network of resistors and sources: charge is not lost at a node, and energy is not gained on a closed path. Together with Ohm's law they give as many equations as there are unknown currents. For the small networks of this lecture we mostly combine resistors in series and in parallel step by step.

Kirchhoff's current law (KCL) and voltage law (KVL)



- KCL: at every node the sum of incoming currents equals the sum of outgoing currents: $I_1 + I_2 = I_3 + I_4$
- KVL: around every closed loop the sum of voltages is zero: $V_0 = V_1 + V_2$

Series and parallel connection

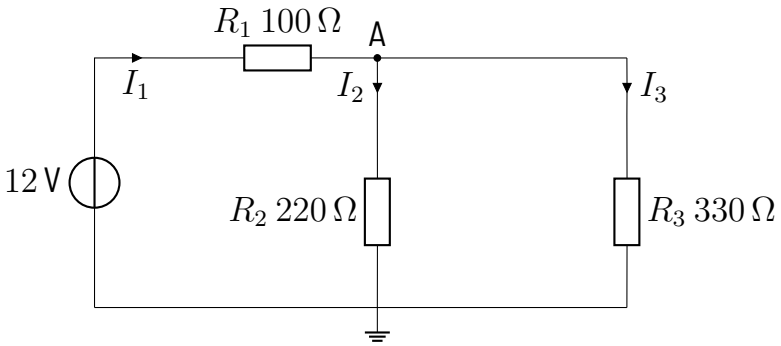


Series and parallel

series: $R = R_1 + R_2$ parallel: $R = R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

- series: same current, the voltages add up
- parallel: same voltage, the currents add up
- parallel resistance is always smaller than the smallest resistor; $R \parallel R = R/2$

Example: series-parallel network



- $R_2 \parallel R_3 = 132\,\Omega$, total $R = 232\,\Omega$, $I_1 = 12\,\text{V}/232\,\Omega = 51.7\,\text{mA}$
- $V_A = I_1 \cdot 132\,\Omega = 6.83\,\text{V}$; $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ (KCL: $I_1 = I_2 + I_3$)

▶ Falstad: `examples/basics/network.txt` The procedure is always the same: combine the parallel resistors, add the series resistor, calculate the total current, then go back and calculate the node voltage and the branch currents. KCL at node A is a cheap check of the result. Exercise 1.1 looks at the power of each resistor.

Exercise 1.1: Power in a series-parallel network

Use the network above ($12\,\text{V}$, $R_1 = 100\,\Omega$, $R_2 = 220\,\Omega$, $R_3 = 330\,\Omega$). All resistors are $0.25\,\text{W}$ types.

a) Calculate the power dissipated in each resistor and the power delivered by the source.

b) Which resistor is overloaded? Which rating would you choose instead?

c) Check the currents and powers in Falstad.

Solution 1.1

a) $P_1 = I_1^2 R_1 = (51.7\,\text{mA})^2 \cdot 100\,\Omega = 268\,\text{mW}$, $P_2 = V_A^2/R_2 = (6.83\,\text{V})^2/220\,\Omega = 212\,\text{mW}$, $P_3 = (6.83\,\text{V})^2/330\,\Omega = 141\,\text{mW}$. Source: $P = 12\,\text{V} \cdot 51.7\,\text{mA} = 621\,\text{mW} = P_1 + P_2 + P_3$.

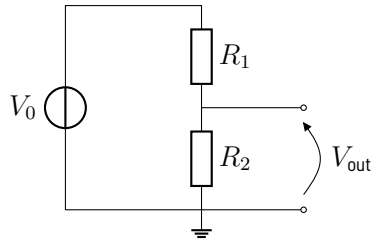
b) R_1 exceeds $0.25\,\text{W}$; R_2 is at $85\,\%$ of its rating, which is too hot as well. With the $50\,\%$ rule R_1 needs $0.6\,\text{W}$ and R_2 $0.5\,\text{W}$ (better: $1\,\text{W}$ for R_1).

c) ▶ Falstad: `examples/basics/network.txt`: `VA` $= 6.83\,\text{V}$; hovering over the resistors shows $I_1 = 51.7\,\text{mA}$, $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ and the power of each resistor.

1.3 The Voltage Divider

The voltage divider is the most frequently used circuit in this lecture. It sets operating points, scales signals, forms the reference ladder of a flash converter and, with a sensor in one branch, it is half of a measurement bridge. Its weak point is the load: as soon as current is drawn from the tap, the output voltage drops.

Unloaded divider



- the same current flows through R_1 and R_2: $I = V_0/(R_1 + R_2)$
- the voltages divide in the ratio of the resistances

Voltage divider

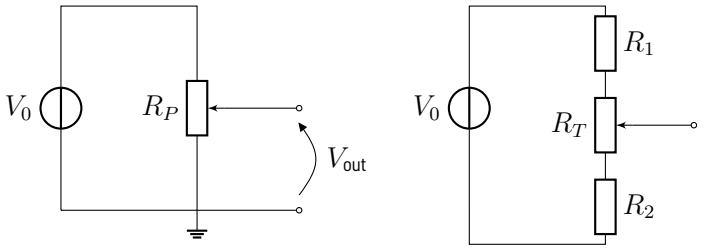
$$V_{\text{out}} = V_0 \frac{R_2}{R_1 + R_2}$$

Loaded divider

- a load R_L appears in parallel to R_2: $V_{\text{out}} = V_0 \dfrac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L}$
- seen from the load the divider is a real source with $V_{\text{th}} = V_0\, R_2/(R_1 + R_2)$ and $R_{\text{th}} = R_1 \parallel R_2$
- rule of thumb: $R_L \geq 10\,(R_1 \parallel R_2)$ keeps the error below $10\,\%$
- for an error below $1\,\%$: $R_L \geq 100\,(R_1 \parallel R_2)$, or buffer the tap with an op-amp

▶ Falstad: `examples/basics/divider_loaded.txt` Choosing the resistance level of a divider is a trade-off. Small resistors make the divider stiff against a load, but they draw a large cross current from the supply and dissipate power. Large resistors save current, but every load, every leakage current and the input current of the next stage disturbs the output. In the flash ADC of the capstone task the reference ladder uses $8 \times 1\,\text{k}\Omega$ at $4\,\text{V}$ ($0.5\,\text{mA}$); it is loaded only by comparator inputs, which draw practically no current.

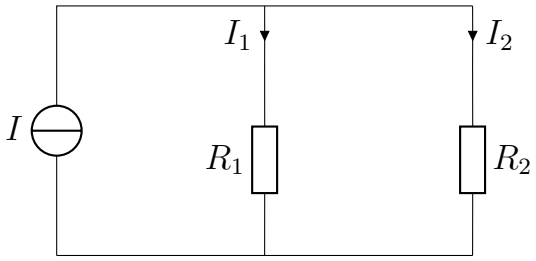
Potentiometer: adjustable divider



- the wiper divides the track: $V_{\text{out}} = V_0 \cdot x$, $x = 0 \ldots 1$ (linear taper)
- trimmer between two fixed resistors: fine adjustment over a small range
- the wiper must not carry a large current; load it lightly

A potentiometer is a resistive track with a sliding contact. Panel potentiometers have a knob, trimmers are adjusted once with a screwdriver. In the capstone task a trimmer sets the gain of the instrumentation amplifier so that the full temperature range gives exactly $0\,\text{V} \ldots 4.00\,\text{V}$. Putting the trimmer between two fixed resistors limits the adjustment range and makes the adjustment much finer: a $1\,\text{k}\Omega$ trimmer between two $4.7\,\text{k}\Omega$ resistors at $10\,\text{V}$ covers only $4.5\,\text{V} \ldots 5.5\,\text{V}$. In Falstad a potentiometer is found under *Draw* → *Passive Components* → *Potentiometer*; it gets a slider automatically.

Current divider



Current divider

$$I_1 = I\,\frac{R_2}{R_1 + R_2} \qquad I_2 = I\,\frac{R_1}{R_1 + R_2}$$

- the **smaller** resistor takes the **larger** current
- example: $10\,\text{mA}$ into $1\,\text{k}\Omega \parallel 4.7\,\text{k}\Omega$: $I_1 = 8.25\,\text{mA}$, $I_2 = 1.75\,\text{mA}$

▶ Falstad: `examples/basics/current_divider.txt`

Exercise 1.2: Loaded voltage divider

A divider with $R_1 = R_2 = 10\,\text{k}\Omega$ is supplied with $V_0 = 10\,\text{V}$.

a) Calculate V_{out} without load.

b) Calculate V_{out} with a load $R_L = 10\,\text{k}\Omega$.

c) Check both results in Falstad.

Solution 1.2

a) $V_{\text{out}} = 10\,\text{V} \cdot \frac{10}{10+10} = 5\,\text{V}$.

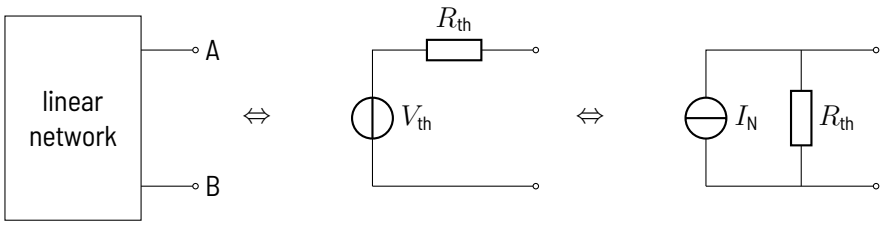
b) $R_2 \parallel R_L = 5\,\text{k}\Omega$, so $V_{\text{out}} = 10\,\text{V} \cdot \frac{5}{10+5} = 3.33\,\text{V}$. The load costs one third of the output voltage, because R_L is only twice the source resistance $R_{\text{th}} = 5\,\text{k}\Omega$.

c) ▶ Falstad: `examples/basics/divider_loaded.txt`
the labels show `Vout_open` $= 5.00\,\text{V}$ and `Vout_load` $= 3.33\,\text{V}$ (move the mouse over the label).

1.4 Equivalent Sources and Superposition

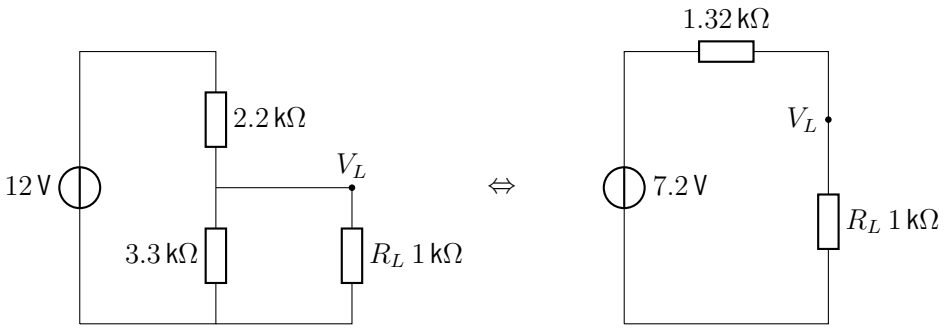
Every linear network of sources and resistors behaves at two terminals exactly like one ideal voltage source with one series resistor (Thevenin) or like one ideal current source with one parallel resistor (Norton). This is the most useful simplification in circuit design: a bridge half, a sensor with its supply or a divider are replaced by one source and one resistor, and the effect of the next stage (the load) can be calculated in one line. The superposition principle is the second tool for linear networks with several sources.

Thevenin and Norton equivalent



- open-circuit voltage: $V_{th} = V_{oc}$ (nothing connected to A–B)
- short-circuit current: $I_N = I_{sc}$ (A and B shorted)
- internal resistance $R_{th} = V_{oc}/I_{sc}$, or: set all sources to zero (voltage source $\rightarrow$ wire, current source $\rightarrow$ open) and calculate the resistance between A and B

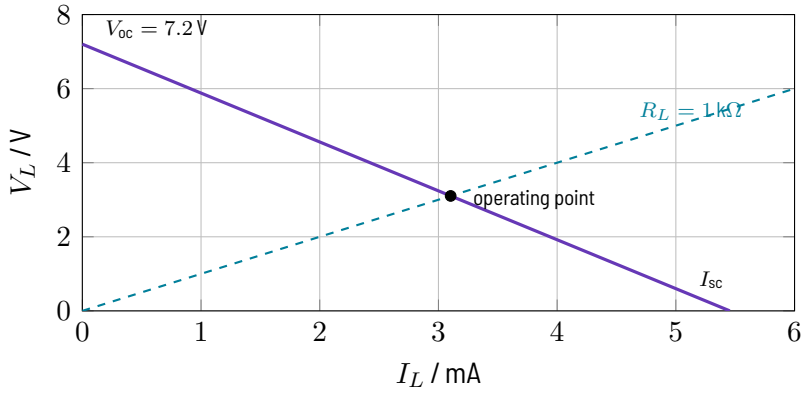
Example: Thevenin equivalent of a divider



- $V_{oc} = 12\,\text{V} \cdot 3.3/5.5 = 7.2\,\text{V}, \quad I_{sc} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$
- $R_{th} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega = 2.2\,\text{k}\Omega \parallel 3.3\,\text{k}\Omega$
- with R_L: $V_L = 7.2\,\text{V} \cdot 1/(1 + 1.32) = 3.10\,\text{V}$ in both circuits

▶ Falstad: `examples/basics/thevenin.txt`

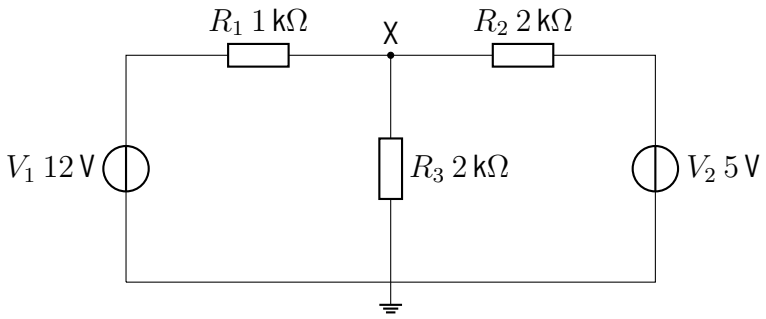
Load line of a real source



- the source can deliver every point on the line $V_L = V_{th} - R_{th}\, I_L$
- the load forces $V_L = R_L\, I_L$; the intersection is the operating point

The load line is the graphical form of the Thevenin equivalent. It connects the open-circuit voltage on the voltage axis with the short-circuit current on the current axis. Its slope is $-R_{th}$: the flatter the line, the stiffer the source. The method also works for nonlinear loads (an LED, a Zener diode): draw the characteristic of the load into the same diagram and read the intersection. The next chapter uses it for real voltage sources.

Superposition



- in a linear network the effect of all sources is the sum of the effects of each source alone
- the other sources are set to zero: voltage source → wire, current source → open
- V_1 alone: $V_{X1} = 12\,\text{V} \cdot \frac{R_2 \parallel R_3}{R_1 + R_2 \parallel R_3} = 6\,\text{V}$; V_2 alone: $V_{X2} = 5\,\text{V} \cdot \frac{R_1 \parallel R_3}{R_2 + R_1 \parallel R_3} = 1.25\,\text{V}$
- both: $V_X = 7.25\,\text{V}$ ▶ Falstad: `examples/basics/superposition.txt`

Superposition is valid only for linear circuits: it fails as soon as a diode switches or an op-amp clips. It is also not valid for power, because power is a quadratic function of current. In the lecture it is used for amplifiers with several inputs, for example the summing amplifier or an op-amp circuit with an offset voltage at one input.

Exercise 1.3: Thevenin equivalent from open-circuit voltage and short-circuit current

The divider $12\,\text{V}$, $R_1 = 2.2\,\text{k}\Omega$ (top), $R_2 = 3.3\,\text{k}\Omega$ (bottom) feeds a load $R_L = 1\,\text{k}\Omega$.

a) Determine V_{oc} and I_{sc} at the tap and from them R_{th}.

b) Calculate V_L with the equivalent circuit.

c) Which load would reduce the open-circuit voltage by only $5\,\%$?

d) Compare the original circuit and the equivalent in Falstad.

Solution 1.3

a) $V_{oc} = 12\,\text{V} \cdot 3.3/(2.2+3.3) = 7.2\,\text{V}$. With the tap shorted no current flows in R_2: $I_{sc} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$. $R_{th} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega$.

b) $V_L = 7.2\,\text{V} \cdot \frac{1}{1+1.32} = 3.10\,\text{V}$.

c) $V_L/V_{oc} = R_L/(R_L + R_{th}) = 0.95 \Rightarrow R_L = 19\,R_{th} = 25.1\,\text{k}\Omega$ (E12: $27\,\text{k}\Omega$).

d) ▶ Falstad: `examples/basics/thevenin.txt`: `VL_orig` = `VL_equiv` = $3.10\,\text{V}$. Replace R_L by a wire and hover over it: the short-circuit current is $5.45\,\text{mA}$ in both circuits.

Exercise 1.4: Superposition

Use the circuit of the superposition slide.

a) Calculate V_X with superposition.

b) Check the result with KCL at node X.

c) How large is the current through R_2, and in which direction does it flow?

d) Check the three cases in Falstad.

Solution 1.4

a) V_1 alone: $R_2 \parallel R_3 = 1\,\text{k}\Omega$, $V_{X1} = 12\,\text{V} \cdot 1/2 = 6\,\text{V}$. V_2 alone: $R_1 \parallel R_3 = 0.667\,\text{k}\Omega$, $V_{X2} = 5\,\text{V} \cdot 0.667/2.667 = 1.25\,\text{V}$. $V_X = 7.25\,\text{V}$.

b) $\frac{12-7.25}{1} = 4.75\,\text{mA}$ flows in through R_1; $\frac{7.25}{2} = 3.625\,\text{mA}$ flows out through R_3 and $\frac{7.25-5}{2} = 1.125\,\text{mA}$ through R_2: $4.75 = 3.625 + 1.125$ ✓

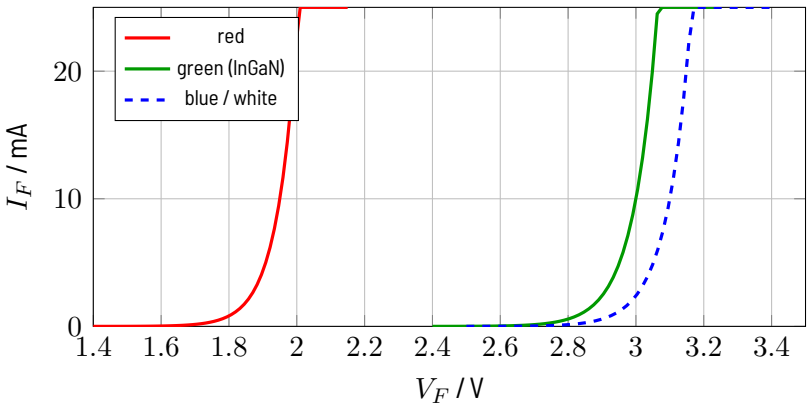
c) $1.125\,\text{mA}$ from X into the $5\,\text{V}$ source: V_2 is charged (absorbs power).

d) ▶ Falstad: `examples/basics/superposition.txt`: `VX1` $= 6.00\,\text{V}$, `VX2` $= 1.25\,\text{V}$, `VX` $= 7.25\,\text{V}$.

1.5 The LED with Series Resistor

The light emitting diode is the first nonlinear component of the lecture and the first task of the lab. It is also the last element of the capstone chain: the digital value of the flash converter is shown on LEDs, "exactly the LED and resistor from Task 1". An LED must never be connected directly to a voltage source. Its current rises exponentially with the voltage, so a few tenths of a volt make the difference between dark and destroyed. The series resistor turns the voltage source into a (rough) current source and sets the current.

LED characteristic



- below a threshold almost no current, above it the current rises steeply
- simple model: constant forward voltage V_F at the operating current

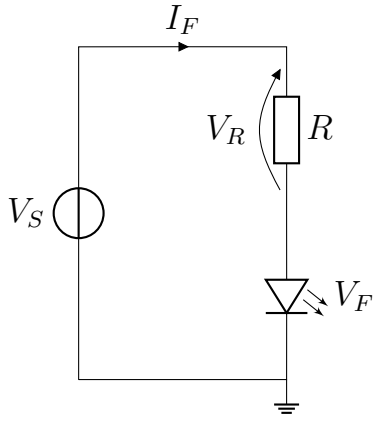
Forward voltage and current

colour	material	V_F at 10 mA … 20 mA
infrared	GaAs	1.2 V … 1.5 V
red	AlGaInP	1.8 V … 2.1 V
yellow, orange	AlGaInP	2.0 V … 2.2 V
green (classic, yellow-green)	GaP	2.1 V … 2.4 V
green (true green), blue, white	InGaN	2.8 V … 3.4 V

- standard 3 mm/5 mm LEDs: $I_F \leq 20$ mA; modern LEDs are clearly visible at 2 mA … 5 mA
- reverse voltage only about 5 V: an LED is not a rectifier

The forward voltage grows with the photon energy, i.e. it is higher for shorter wavelengths: red needs less than 2 V, blue and white (a blue LED with a phosphor) need about 3 V. This matters for a 3.3 V supply, where a blue LED leaves only a few hundred millivolts for the series resistor and the current depends strongly on the spread of V_F. The forward voltage also decreases by about 2 mV/K, so an LED without a resistor heats up, draws more current and heats up further.

Dimensioning the series resistor



LED series resistor

$$R = \frac{V_S - V_F}{I_F} \qquad P_R = (V_S - V_F) \cdot I_F$$

- example: $V_S = 5\,\text{V}$, red LED $V_F = 2.0\,\text{V}$, $I_F = 10\,\text{mA}$: $R = 300\,\Omega \Rightarrow$ E12: $330\,\Omega$, $I_F = 9.1\,\text{mA}$, $P_R = 27\,\text{mW}$

Design recipe

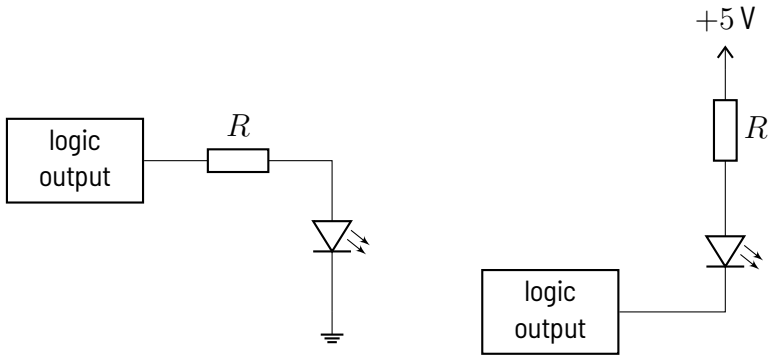
Design card: LED with series resistor

1. take V_F from the datasheet at the desired current (or the table: red $2.0\,\text{V}$, blue/white $3.0\,\text{V}$)
2. choose I_F: $2\,\text{mA}$ … $10\,\text{mA}$ for an indicator, check the maximum current of the driver
3. $R = (V_S - V_F)/I_F$, round **up** to the next E12/E24 value
4. recalculate I_F and check $P_R = I_F^2 R$ against the rating

- rounding up keeps the current below the target, which protects LED and driver
- the larger $V_S - V_F$, the less I_F depends on the spread of V_F

The last point explains why the resistor works as a current source: the current is $I_F = (V_S - V_F)/R$. If V_F changes by $0.1\,\text{V}$, the current changes by $0.1\,\text{V}/R$, which is $3\,\%$ at $V_S = 5\,\text{V}$ but $8\,\%$ at $3.3\,\text{V}$ for a red LED. For a blue LED at $3.3\,\text{V}$ the scheme hardly works any more; a real current source (chapter Sources) is better.

Driving an LED from a logic output



source: high = on sink: low = on

- a logic output is a switch to V_S or to ground with an output resistance of some $10\,\Omega$
- 74HC at $5\,\text{V}$: $\pm 4\,\text{mA}$ with guaranteed levels, absolute maximum $25\,\text{mA}$ per pin
- microcontrollers: some mA per pin and a limit for the whole port
- $330\,\Omega$ at $5\,\text{V}$ gives about $9\,\text{mA}$, $1\,\text{k}\Omega$ about $3\,\text{mA}$

▶ Falstad: `examples/basics/led_resistor.txt` A CMOS output stage consists of a p-channel transistor to the supply and an n-channel transistor to ground. Each acts like a resistor of some $10\,\Omega$ to $50\,\Omega$ when it is switched on. With an LED current of a few milliampere the output voltage therefore differs by only $0.1\,\text{V}$ … $0.3\,\text{V}$ from the rail, and the simple formula still applies. The LED can be connected to ground

(the output *sources* the current, the LED is on at high level) or to the supply (the output *sinks* the current, the LED is on at low level). Older TTL outputs can sink much more current than they can source; that is why many datasheets and old circuits connect LEDs to the supply. In the capstone task the three bits of the encoder drive three LEDs to ground. The Falstad example uses ideal logic outputs, so the full $5\,\mathrm{V}$ is available.

Exercise 1.5: LED on a $5\,\mathrm{V}$ logic output

A red LED ($V_F = 2.0\,\mathrm{V}$ at $10\,\mathrm{mA}$) is to be driven by a $5\,\mathrm{V}$ logic output.

a) Calculate the series resistor for $10\,\mathrm{mA}$ and choose an E12 value.

b) Calculate the actual current and the power in the resistor.

c) The same LED is to be driven from a $3.3\,\mathrm{V}$ microcontroller pin with at most $4\,\mathrm{mA}$. Choose the resistor.

d) Falstad uses an LED model with $V_F \approx 1.8\,\mathrm{V}$. Which current does the simulation show with $330\,\Omega$?

Solution 1.5

a) $R = (5\,\mathrm{V} - 2.0\,\mathrm{V})/10\,\mathrm{mA} = 300\,\Omega$; rounded up to E12: $330\,\Omega$.

b) $I_F = 3.0\,\mathrm{V}/330\,\Omega = 9.1\,\mathrm{mA}$, $P_R = 3.0\,\mathrm{V} \cdot 9.1\,\mathrm{mA} = 27\,\mathrm{mW}$: any resistor will do, even SMD 0402.

c) $R \geq (3.3\,\mathrm{V} - 2.0\,\mathrm{V})/4\,\mathrm{mA} = 325\,\Omega$, again $330\,\Omega$ ($I_F = 3.9\,\mathrm{mA}$; V_F is a bit lower at this current, so the real current is slightly higher).

d) ▶ Falstad: `examples/basics/led_resistor.txt`: $V_{LED} = 1.78\,\mathrm{V}$, so $I_F = (5 - 1.78)\,\mathrm{V}/330\,\Omega = 9.76\,\mathrm{mA}$ (hover over the resistor). Use the slider to see how the current changes with R.

1.6 Capacitor and RC Circuits

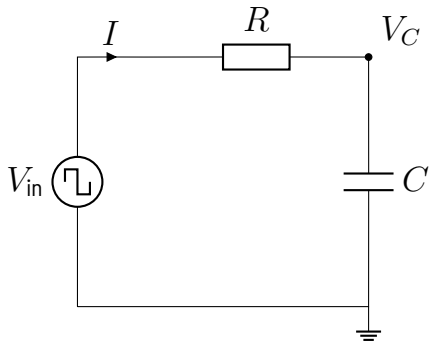
Capacitors store charge. In this lecture they appear as decoupling capacitors at every supply pin, in filters, in the hold capacitor of the sample and hold circuit and as the reason for every delay. The RC circuit is the simplest dynamic circuit; its exponential response and its time constant $\tau = RC$ appear again and again.

The capacitor

- charge proportional to voltage: $Q = C \cdot V$
- current = change of charge: $I = C \dfrac{dV}{dt}$
  - the voltage across a capacitor cannot jump (that would need an infinite current)
  - DC: once charged, no current flows (open circuit)
- energy: $E = \frac{1}{2} C V^2$
- types: ceramic (pF to 100 µF; C0G stable, X7R voltage dependent), film (precise, for filters and hold capacitors), electrolytic (large, polarized)

The relation $I = C\,dV/dt$ is the capacitor's counterpart of Ohm's law. A constant current charges a capacitor linearly: 1 mA into 1 µF gives 1 V/ms. This is used in integrators and ramp generators. Ceramic capacitors of type X7R or Y5V lose a large part of their capacitance with DC voltage and temperature; for filters and the hold capacitor of a sample and hold circuit use C0G (NP0) ceramic or film capacitors.

Charging through a resistor

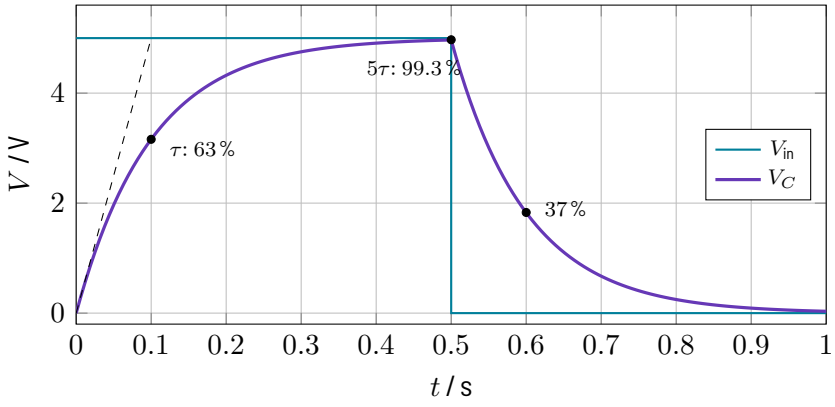


RC charging and discharging

$V_C(t) = V_0\left(1 - e^{-t/\tau}\right) \qquad V_C(t) = V_0\, e^{-t/\tau} \qquad \tau = R\,C$

- step of V_0 at the input; the current $I = (V_{in} - V_C)/R$ decreases as V_C rises
- time to reach a voltage V_1: $t = \tau \ln \frac{V_0}{V_0 - V_1}$

The time constant

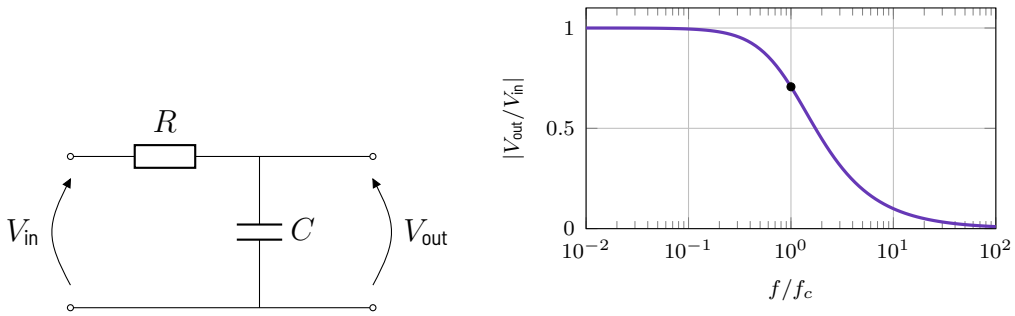


- after τ: 63 %, after 3τ: 95 %, after 5τ: 99.3 % ("settled")

- the tangent at $t = 0$ reaches the final value after exactly τ
- example $R = 10\,\text{k}\Omega$, $C = 10\,\mu\text{F}$: $\tau = 0.1\,\text{s}$ ▶ Falstad: `examples/basics/rc_charging.txt`

How long one has to wait depends on the required accuracy. Settling to within $1\,\%$ takes $4.6\,\tau$, to within half an LSB of an n-bit converter $\tau \ln(2^{n+1})$: $2.8\,\tau$ for the 3-bit converter of the capstone task, but $9\,\tau$ for a 12-bit converter. This is the reason why the acquisition time of a sample and hold circuit is specified together with an accuracy.

Preview: the RC low-pass



Cut-off frequency

$$f_c = \frac{1}{2\pi RC} \qquad \left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \frac{1}{\sqrt{1 + (f/f_c)^2}}$$

- at f_c the amplitude drops to $1/\sqrt{2} = 0.707$ (-3 dB), phase $-45°$
- example: $1.6\,\text{k}\Omega$, $100\,\text{nF}$: $f_c = 995\,\text{Hz}$ ▶ Falstad: `examples/basics/rc_lowpass.txt`

For slow signals the capacitor follows the input; for fast signals it cannot charge in time and the output amplitude decreases. Above f_c the amplitude falls by a factor of 10 per decade of frequency (-20 dB per decade). In the capstone task an anti-aliasing low-pass with $f_c \approx 2.5\,\text{Hz}$ precedes the sample and hold circuit that samples at $5\,\text{Hz}$; with $C = 10\,\mu\text{F}$ this needs $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot 10\,\mu\text{F}) = 6.4\,\text{k}\Omega$, E24: $6.2\,\text{k}\Omega$ ($f_c = 2.57\,\text{Hz}$). The chapter on active filters treats filters in detail. The RC low-pass in the Falstad example is driven with $1\,\text{kHz}$, almost exactly f_c: the output amplitude is $0.705\,\text{V}$ for an input amplitude of $1\,\text{V}$, and the output lags by about a quarter period.

Exercise 1.6: RC charging

A capacitor $C = 10\,\mu\text{F}$ is charged through $R = 10\,\text{k}\Omega$ from a square wave that switches between $0\,\text{V}$ and $5\,\text{V}$ at $1\,\text{Hz}$.

a) Calculate τ and V_C after τ and after $0.5\,\text{s}$.

b) After which time does V_C reach $4\,\text{V}$?

c) To which voltage does C discharge $0.1\,\text{s}$ after the falling edge?

d) Read the values from the Falstad scope.

Solution 1.6

a) $\tau = 10\,\text{k}\Omega \cdot 10\,\mu\text{F} = 0.1\,\text{s}$. $V_C(\tau) = 5\,\text{V}\,(1 - e^{-1}) = 3.16\,\text{V}$; $V_C(0.5\,\text{s}) = V_C(5\tau) = 5\,\text{V}\,(1 - e^{-5}) = 4.97\,\text{V}$.

b) $t = \tau \ln \frac{5}{5-4} = 0.1\,\text{s} \cdot \ln 5 = 0.161\,\text{s}$.

c) The discharge starts at $4.97\,\text{V}$: $V_C = 4.97\,\text{V} \cdot e^{-1} = 1.83\,\text{V}$.

d) ▶ Falstad: `examples/basics/rc_charging.txt`: the scope shows $3.16\,\text{V}$ at $0.1\,\text{s}$, $4\,\text{V}$ at $0.161\,\text{s}$, $4.97\,\text{V}$ at the falling edge and $1.83\,\text{V}$ $0.1\,\text{s}$ later.

1.7 Working with Falstad

Falstad's circuit simulator (CircuitJS) runs in the browser and needs no installation [11]. It simulates the circuit continuously and shows the voltages as colours and the currents as moving dots, which makes it ideal for getting a feeling for a circuit. It is not a replacement for SPICE: its models are simple (the op-amp has no bandwidth limit by default, transistors have no capacitances) and it cannot print exact plots. For the circuits of this lecture it is accurate enough, and every example has been checked against the hand calculation.

Opening and building circuits

- https://www.falstad.com/circuit/circuitjs.html
- open an example: click its Falstad box in the script, or *File* → *Import From Text* and paste the content of `examples/basics/*.txt`
- new circuit: *Circuits* → *Blank Circuit*
- place elements: *Draw* menu or right click; frequent ones have shortcuts (`r` resistor, `c` capacitor, `w` wire, `g` ground)
- drag to place, double click to edit values (`4.7k`, `100n`, `10u` are accepted)
- save your work: *File* → *Export As Text* or *Export As Link*

Pitfalls when drawing

- wires connect **only at their end points**: a wire passing over a terminal is not connected; split the wire there
- unconnected terminals are shown as red dots
- every circuit needs a ground (or a source that defines the potential)
- *labeled nodes* (*Draw* → *Wiring* → *Labeled Node*) with the same name are connected without a wire, and their name can be used for scopes
- "singular matrix": usually a floating node or two ideal voltage sources in parallel

Most errors in self-drawn circuits come from the first rule. A wire that runs over the end of a resistor looks connected, but it is not. Check the dots: a junction of three or more wires is drawn with a large dot. Moving an element with the mouse shows which wires follow it.

Measuring

- hover the mouse over an element: the bottom right corner shows voltage, current and power
- hover over a labeled node: its voltage to ground
- voltage colours: green positive, red negative, grey zero; yellow dots: current
- scope: right click on an element → *View in New Scope*; right click on a scope for settings (scale, several traces, *Max Scale*)
- sliders: *Edit* dialog of an element, check *Show slider*; the examples have sliders for the values worth changing
- simulation speed: slider on the right; the time step is in *Options* → *Other Options*

The examples of this lecture are generated by a Python program that also simulates them and asserts the numbers quoted in this script. When you open an example, compare the displayed values with the solution of the exercise. If they differ, check the time: circuits with capacitors need a few time constants before they reach their final state.
Example: open `led_resistor`. Hover over the label `VLED` to read the forward voltage (1.78 V), hover over the resistor to read the current (9.76 mA), and move the slider to change the resistor. The two LEDs on the right are driven by inverters and blink in turn: one is connected as a current source, one as a current sink.

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